Exploring the Landscape of Cloud Robotics: A Comprehensive Review

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2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT) | 979-8-3503-7024-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCCNT61001.2024.10724868

***Abstract*— Cloud robotics is an innovative field that leverages cloud technologies—including cloud computing (CC), cloud storage, deep learning, big data, and the Internet of Things to augment the capabilities of robotics. This integration facilitates the execution of robotic functions through a converged infrastructure and shared services. The domain faces numerous challenges, such as the increasing demand for cloud security, ensuring performance accuracy, meeting the real-time requirements of diverse scenarios, ensuring rapid response, managing remoteness, and addressing network constraints. Although various studies have conducted systematic literature reviews (SLRs) on aspects like requirements elicitation, modeling, stakeholder analysis, and the selection and prioritization of robotics requirements, there has been limited focus on the technologies, research issues, and applications specific to cloud robotics. Addressing this gap, our paper presents an SLR on the achievements and challenges in cloud robotics, aiming to identify research gaps for future investigations in this field.**

Keywords— robotics, cloud robotics, cloud computing, selection, systematic literature review.

I. INTRODUCTION

In the rapidly evolving landscape where digitization is becoming ubiquitous, industries are transitioning towards Industry 4.0. Within this framework, there is a significant emphasis on machinery automation, reducing reliance on human labor as machines take over complex and tedious tasks. This is achieved by integrating computers within the machines themselves. Given the vast amounts of data involved, machinery requires substantial storage space for data and the programs needed to execute various tasks. To circumvent the need for extensive storage capacity in machines/robots, centralized bulk storage is advocated. Machines/robots connect to this centralized storage via the internet, a system often implemented using cloud technology. James Kuffner of Carnegie Mellon University introduced the term "cloud robotics" in 2010, merging robotics with CC to endow robots with an 'extended and shared brain.' This concept pertains to any robot or automation system that leverages data or code from a network for its operations, thus not confined to standalone systems for sensing, computation, and memory [Homes and Meyer 2010]. At present, 'cloud robotics' (CR) allows robots to utilize computing, storage, and communication resources provided by the cloud. While collaborative robots (cobots) are generally more expensive and have limited capabilities, cloud-connected robots offer cost-efficiency and greater adaptability, facilitating communication and information sharing with other robots through cloud access. Moreover, CC infrastructure for Industry 4.0 is readily accessible to these robots. Recent years have seen a surge in research interest in cloud robotics, evidenced by an increasing volume of scientific publications. However, there has been a noted absence of publications specifically addressing the current state of cloud robotics within Industry 4.0. Aissam's paper titled 'Cloud Robotics and Industry 4.0' [S21] indicates that the high costs associated with implementing Industry 4.0 can be mitigated through cloud robotics. Despite these insights, further research is essential to fully understand cloud robotics' evolution and current trends in Industry 4.0. According to Kumar et al., Industry 4.0 encompasses the automation and control of production systems using a variety of technologies, including information technology and robotics [17]. A collection of emerging technologies, including the Internet of Things (IoT), CC, big data analytics, blockchain, cybersecurity, augmented reality (AR), automation, industrial robots, 3D printing (additive manufacturing), simulation modeling, cyber-physical systems, semantic technologies, real-time communication, advanced computing, and information technology (IT), are seen by many academics as essential building blocks for Industry 4.0.

These findings motivate further exploration in the field of cloud robotics, aiming to identify and evaluate current scenarios of CR in Industry 4.0. This involves conducting a SLR to uncover potential research areas, opportunities, and gaps. The study identifies key technologies like IoT, IoRT, CPS, mobile cloud, AI, and deep learning as crucial for future advancements, while emphasizing security as a top concern requiring further research. Our research reveals a scarcity of studies addressing the presence and impact of cloud robotics techniques on robot selection. This paper presents an SLR on cloud robotics techniques for robot selection, following the guidelines of Kitchenham and Charters [1].

The paper's organization is as follows: Section II presents the literature review; Section III explores CC and its service models; Section IV outlines potential threats to validity; Section V summarizes the findings of our SLR and addresses the research questions; Section VI examines unresolved issues, challenges, and future research directions; and finally, Section VII offers conclusions and proposes potential paths for future research endeavors.

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II. LITERATURE REVIEW

Studies [S1], [S22], [S20], and [S21] highlight the cloud's role in enhancing robotics and automation systems through big data, CC, collective robot learning, and human computation. OpenEASE, as demonstrated in [S2], exemplifies knowledge exchange between robots, emphasizing the cloud's facilitation of knowledge sharing and task adaptation.

Research [S3] introduces frameworks for robotic collaboration within the cloud, focusing on task cooperation and knowledge sharing, while identifying key challenges. Similarly, [S4] and [S15] present cloud services and frameworks that address central control, robot intelligence, privacy, and security concerns, alongside a mechanism for knowledge fusion and transfer among robots.

Studies [S5], [S11], [S12], and [S34] explore multi-robot path planning, task allocation in cloud manufacturing, and the integration of manufacturing systems with cloud services. These works propose algorithms and frameworks to improve efficiency and selection in manufacturing environments.

The green software model developed in [S8] and the fuzzy-rule-based decision process in [S9] aim at reducing energy consumption in cloud robotics and improving recycling rates of smart products, respectively.

Research [S10] leverages blockchain technology to facilitate secure and low-cost agreements in cloud manufacturing platforms, underscoring the technology's potential to enhance trust and efficiency.

Studies [S6], [S37], and [S40] discuss the evolution of software platforms for educational purposes, the development of security frameworks, and the legal challenges of integrating cloud robotics within dynamic ecosystems.

Contributions from [S26], [S33], [S35], [S36], and [S38] include cutting-edge architectures for tactile communication, network traffic reduction, improved AGV coordination, and secure cloud robotics infrastructures.

Articles [S16], [S19], [S28], [S32], and [S39] survey current developments, propose novel methodologies, and outline open research issues in cloud robotics, emphasizing the need for further exploration in areas such as quality of service, resource allocation, and security.

III. CLOUD COMPUTING AND SERVICE MODELS

The evolution of CC is thoroughly detailed in academic and industry sources [40]. The term "CC" was first introduced by Google's CEO at a search engine conference in 2006, marking the beginning of a significant shift in how computing resources are utilized and delivered. The term "Cloud" signifies the delivery of services over the internet [41]. Notable advancements that have enabled the growth of CC include the construction of the internet backbone, widespread adoption of wireless internet connections, development of robust server and data center infrastructure, and innovations in high-performance computing technologies.

A report by IDC shows strong growth in CC revenue, with a year-on-year increase of 8.1% to $8.4 billion in Q3 2016. They also predict public cloud spending to outpace overall IT spending by a factor of seven, reaching $203.4 billion globally by 2019.

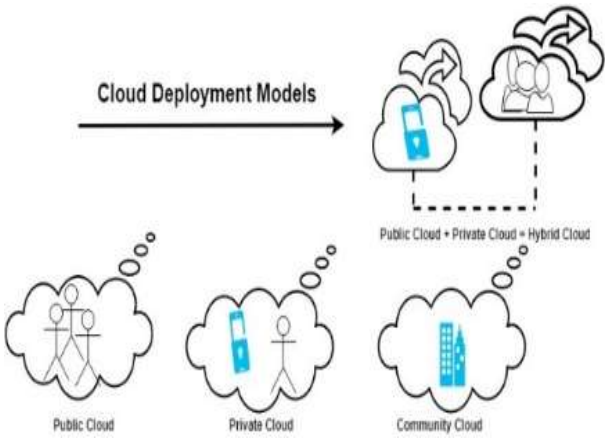


Fig. 1. CC Deployment Models

The market for CC has experienced swift growth and intense competition, with major players like Amazon, Google, and Microsoft continually refining their cloud platforms to cater to a wide range of customers. This success has spurred additional companies such as Media Temple, Mosso, Joyent, and others to venture into the cloud services market.

CC utilizes three key delivery models to offer various services (see Figure 1 and Figure 2)—Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS)—to consumers [42]. SaaS enables users to run applications on a cloud provider's infrastructure, accessible through web browsers. PaaS allows clients to develop, test, and deploy applications using cloud-based resources. IaaS provides access to essential computing resources, such as storage and networking, offering users flexibility in managing applications.

Mell and Grance (2011) describe essential characteristics of CC, including self-service provisioning, universal network access, resource pooling, scalability, and pay-per-use billing

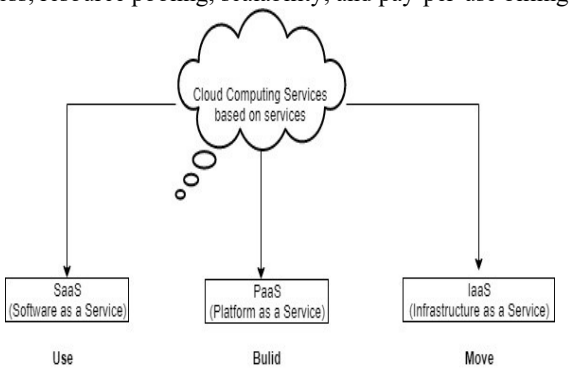


Fig. 2. CC services

[44,45]. These characteristics ensure that users can provision resources without manual intervention, access services from various devices, efficiently share resources among many users, scale services rapidly to meet demand, and have visibility into resource usage for better management [Mell and Grance, 2011].

Cloud service providers offer their services through three primary models: SaaS, PaaS, and IaaS [46, 47]. These models provide a range of CC services to meet various needs, from software distribution to hardware infrastructure. Beyond these, the CC landscape includes a variety of other service models such as Container as a Service (CaaS), Desktop as a

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Service (DaaS), Hardware as a Service (HaaS), and several others, indicating a diverse range of services tailored to specific domains and requirements.

The Cloud Service Model Framework (CSMF) outlines three main cloud service offerings: SaaS delivers applications directly over the web, eliminating installation needs. PaaS provides a platform for users to build and manage their applications, handling the underlying infrastructure. Finally, IaaS offers fundamental resources like storage and servers, giving users the most control over their software environment. These options cater to different needs, from user-friendly applications to customizable infrastructure.

IV. METHODOLOGY

To identify gaps in existing research on cloud robotics techniques related to robot selection and to outline potential areas for future investigation, we have followed the methodology outlined by Kitchenham and Charters [1]. This methodology involves defining research questions (RQs), devising a search strategy, conducting study selection, and synthesizing the gathered data. Our review approach is based on these procedural steps.

A RQs

The SLR is performed to identify the application of cloud robotics techniques for robot selection. The following RQs were created to help us reach this goal:

- *RQ-1:* What are the different technologies, research themes, and applications used in cloud robotics?
- *RQ-2:* What is the relationship between MaaS infrastructure and individual industrial plants in which a robot may be empowered through a cloud environment?
- *RQ-3:* How can cloud robotics aid in the process of identifying robots during requirement selection?
- *RQ-4:* What are the methods for implementing automation in cloud robotics within present-day contexts?

B Search strategy

The main studies were chosen from five significant electronic databases, namely, "IEEE Xplore, ACM Digital Library, Springer, Science Direct, and Google Scholar." We formulated the search query outlined below based on the previously mentioned RQs.

Search string: "*((Cloud computing OR robotics OR cloud robotics selection OR requirements) AND (system OR robotics OR constraints OR shortcomings OR strong point OR drawbacks) AND (survey OR SLR OR literature review OR journey OR mapping OR SLR mapping OR state-of-the-art)).*"

C Study selection

Using the search criteria outlined in section 2 (B), we have chosen primary studies published between 2009 and 2024. Figure 3 depicts the overview of the selection process. In the initial stage, we selected 170 studies from databases such as "Springer", "ACM Digital Library", "Science Direct", and "IEEE Xplore", "Google Scholar".

Following an evaluation based on their titles, several of the initially chosen studies were found to be redundant and superfluous and were consequently removed from the SLR. Following that, 110 primary studies were pinpointed. These chosen papers were subjected to additional evaluation based on their abstracts and conclusions, leading to the selection of 60 studies for further examination. Quality assessment (QA) criteria were subsequently utilized in subsequent selection stages to assess the chosen studies. Following this process, 40 studies were identified and selected. These primary studies serve as the basis for addressing the formulated RQs.

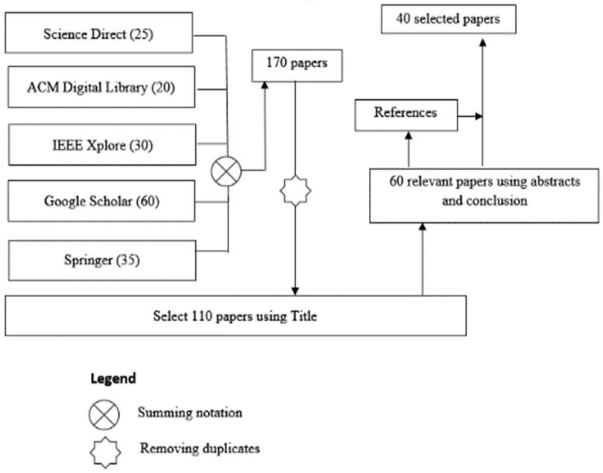


Fig. 3. Search methodology employed during the process of selecting studies

To ensure the quality of included studies in this review, we applied specific criteria. These criteria evaluated the studies' relevance to the research questions (QA-1), clarity of research objectives (QA-2), the presence of supporting case studies (QA-3), and their contribution to either industry or academia (QA-4). Studies that partially met a criterion received 0.5 points, while those fully meeting it received 1 point. Only studies scoring 2 or higher (out of 4 possible) were included as primary references (detailed criteria are available in Appendices A and B).

D. Data synthesis

The information from the 40 primary studies is analyzed to address the research questions formulated. This analysis is conducted through the following methods to synthesize the data:

- The response to RQ-1 is presented in Figure 3, demonstrating the diverse technologies, research topics, and applications employed in CR.
- The analysis for the RQ-2 is the relationship between MaaS infrastructure and individual industrial plants in which a robot may be empowered through a cloud environment.
- The observations to the RQ-3 and RQ-4 are synthesized in textual form.

V. THREATS OF VALADITY

The selection strategies outlined in section 2(B) were employed to identify studies for the systematic literature review (SLR), utilizing a specific search string to find pertinent studies in the databases mentioned. Despite thorough efforts, there's a chance some relevant studies might have been overlooked, particularly those not explicitly

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mentioning terms tied to the research questions established in section 2(A). To address this, manual searches were conducted to capture any studies that might have slipped through the initial screening. Additionally, to ensure the quality of the studies included, each one was independently assessed using a set of four quality assurance (QA) questions.

VI. RESULTS AND DISCUSSION

In the SLR, we have considered the publications from 2011 to 2021 and have identified 40 primary studies as shown in Figure 3. Figure 4 depicts the number of publications over time. The results affirm the presence of research on cloud robotics within the context of Industry 4.0. Although relatively scarce, there has been an upward trend in the number of publications in recent years, indicating a burgeoning interest in this domain.

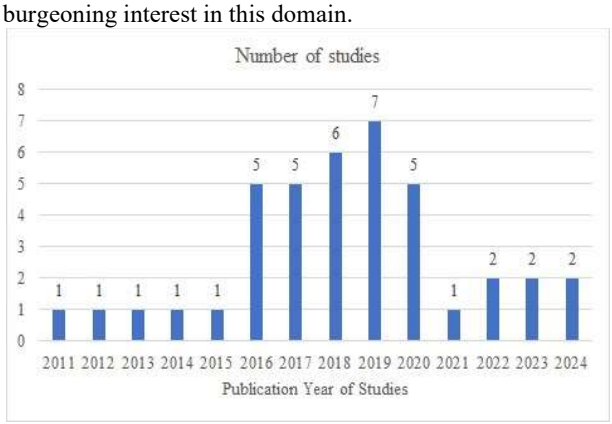


Fig. 4. Yearly Organization of Primary Studies

RQ-1: What are the different technologies, research themes, and applications used in cloud robotics?

Researchers have delved into a range of advanced technologies in their investigations, observing that some are commonly utilized across studies, while others are tailored to specific project objectives or organizational requirements. Noteworthy technologies identified include Big Data, the IoT, the IoRT, cyber-physical systems, mobile cloud, AI, ML, and NN, as cited in multiple studies ([S1], [2], [S23], [S28], [3], [4], [S21], and others). Analysis reveals prominent themes in CR for Industry 4.0, such as ambient intelligence, autonomous robots, AGV, MC, localization and mapping, service robots, Context-Aware Cloud Robotics, and robot swarms, which are emphasized in several publications ([2], [5], [7], [8], [S25], [S28], [S35], [S40], and more). The research also encompasses diverse applications in industrial processes, manufacturing, agriculture, and the oil and gas sectors, demonstrating the wide-ranging impact of these technologies ([4], [12], [13], [16], [S21], [S24], [S34], [S35], [S36], [S40]).

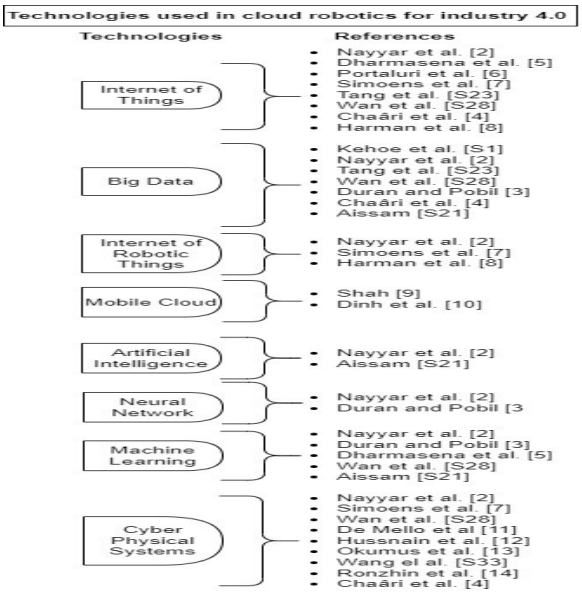


Fig. 5. Technologies used in cloud robotics

RQ-2: What is the relationship between MaaS infrastructure and individual industrial plants in which a robot may be empowered through a cloud environment?

The concept of Manufacturing as a Service (MaaS) leverages cloud technology to connect individual manufacturing plants to a centralized cloud platform, enabling information exchange and integration. This setup allows for industrial robots to be enhanced through cloud connectivity, leading to more sophisticated and interconnected robotic systems.

Robot as a Service (RaaS) capitalizes on the cloud's vast data resources to refine robot interactions and decision-making processes. Major progress in CR is propelled by ideas such as remote operation, harnessing big data to create shared knowledge repositories, collective learning, improved intelligence, and behavioral patterns, and utilizing cloud architecture to enhance robot performance [19]. Cloud-enabled robotics integrates software services, computing devices, sensors, and robots within a cloud framework, structured into three layers [20]:

- The cloud service layer provides robot-accessible application services, facilitating direct interaction with robots.
- A middleware communication and collaboration layer oversee the interaction between cloud services and robots, utilizing a protocol based on message exchange.
- The control layer focuses on the robot itself, including its hardware capabilities and direct control mechanisms, ensuring seamless operation and functionality.

In the cloud service tier, application services are utilized to construct interfaces for robots. Supervising the communication between cloud services and robots is a middleware termed the cloud collaboration framework, functioning on a message-based model.

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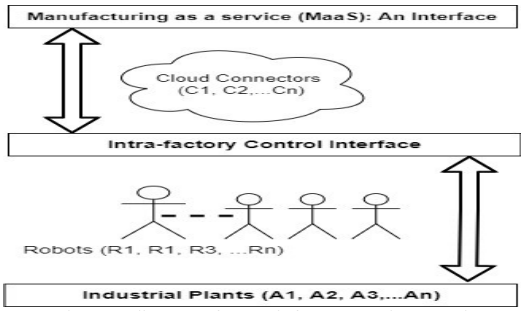


Fig. 6. A diagram of MaaS infrastructure in general

This research proposes a system where a physical robot acts as a computer program element, performing reasoning tasks directly within its control system [20].

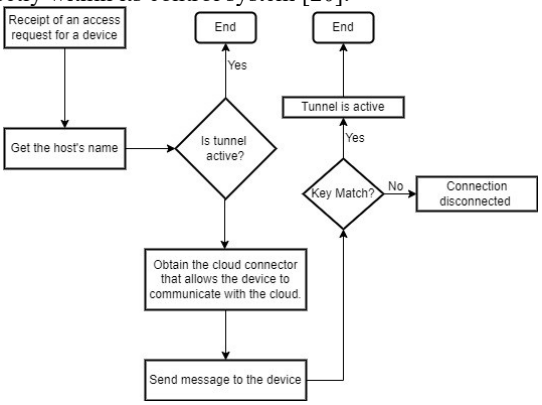


Fig. 7. The process of allowing remote access to a network

Building on the work of [21], Figure 7 depicts how to remotely access a cloud-connected network device. A user interface is requested by a client.

- The device enabled for cloud connectivity is linked to a LAN.
- To request the detection of a cloud connector (CCN), a designated hostname is needed.
- An open relay message is sent by the CCN to a CEN device over a management connection.
- The system creates a secure tunnel between the CR and the cloud-enabled device (CEN device).
- Clients connect securely to the CEN device over the internet.

RQ-3: How can cloud robotics aid in the process of identifying robots during requirement selection?

The architecture for selecting robots must accommodate a variety of robots, regardless of brand or technical specifications, through a unified model. It also needs a system for orchestrating multiple robot services and a method for publishing and discovering robot capabilities, ensuring that services from different IaaS providers can be federated and seamlessly integrated. Key innovations include a robot-as-a-Service platform for easy access to diverse robots [22], a cloud robotics service extension [23], and DAvinCi PaaS for processing robotic data in the cloud [24]. These advancements facilitate data sharing among different robots and offload computational tasks to the cloud, enhancing efficiency. The approach utilizes current technology for the publication and discovery of robot services, with a presence service to quickly identify robots for tasks like search and rescue, optimizing robot selection and deployment [27-28].

RQ-4: What are the methods for implementing automation in cloud robotics within present-day contexts?

Cloud technology enhances robotics and automation by allowing systems to access extensive data and computational resources from the network, moving beyond the limitations of standalone systems. Since 2001, the IEEE Robotics and Automation Society has emphasized the cloud's role in providing resources like images, video, and real-time data that surpass traditional database capacities [29-32]. CC facilitates advanced robotics applications, including navigation and SLAM, by offloading heavy computational tasks to the cloud [33-34]. The cloud supports the growth of cloud robotics, promoting access to datasets, software, and development tools, alongside fostering open-source collaborations, as seen with the Robot Operating System (ROS) [35-38]. Rapyuta, part of the RoboEarth project, showcases the shift towards cloud-based computing, integrating big data and offering a platform that simplifies computing for robots [39]. This evolution towards a SaaS model introduces RaaS, streamlining software, data, and hardware management by processing data on remote servers, significantly simplifying the operational complexity for users.

VII. OPEN QUESTIONS, CHALLENGES, AND PROSPECTS

Since its beginnings in 2011, cloud robotics has progressed rapidly, with numerous studies in literature focusing on its architecture and applications. However, to increase the performance of cloud-based robotic systems, some existing obstacles and issues in cloud robotics systems must be addressed. We will go over some of the open questions and challenges in this part. The primary obstacles and future trends in cloud robotics are depicted in Figure 8.

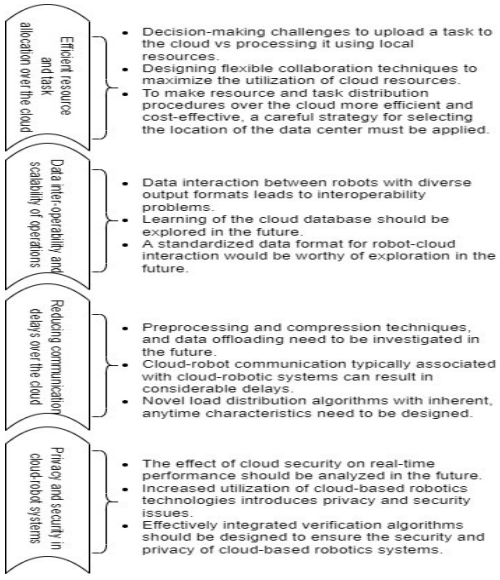


Fig. 8. Obstacles and Prospects in CR

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VIII. CONCLUSION: A CRITICAL ANALYSIS AND DISCUSSION

Robotics, particularly cloud robotics, are evolving swiftly, marking a significant shift from traditional robots with onboard controllers to those managed by a cloud-based "intelligence" residing in data centers. This advancement leverages Artificial Intelligence and cutting-edge software, paving the way for novel value chains, technological breakthroughs, architectures, experiences, and business frameworks. In our study, we delved into the progress and hurdles of cloud robotics through a SLR, examining 40 primary studies to address four key research inquiries. Our investigation reveals a range of technologies, research themes, and applications pivotal to cloud robotics. We explore the synergy between Manufacturing as a Service (MaaS) infrastructures and industrial plants, highlighting the role of cloud environments in enhancing robotic capabilities. Our criteria for selecting cloud robotics architectures include a unified model, efficient publication and discovery mechanisms, a composition framework, and the integration of IaaS for robots. Moreover, we discuss strategies to facilitate cloud robotics automation, drawing on extensive literature reviews. Our findings underscore the need for research on real-time control, improved selection processes for cloud robots, and advancements in video and voice processing to better understand environments and improve communication with stakeholders. This exploration not only highlights current achievements but also charts a course for future research in automating and advancing cloud robotics.

Appendix A: Primary Studies

Paper ID	DOI of the Paper
$P_{id}1$	DOI: 10.1109/TASE.2014.2376492
$P_{id}2$	DOI: 10.1109/LRA.2018.2794626
$P_{id}3$	DOI: 10.1109/ACCESS.2018.2852295
$P_{id}4$	DOI: 10.1109/ACCESS.2018.2865585
$P_{id}5$	DOI: 10.3390/sym11101241
$P_{id}6$	DOI: 10.1016/j.ifacol.2016.11.124
$P_{id}7$	DOI: 10.1016/j.future.2016.01.002
$P_{id}8$	DOI: 10.1016/j.compeleceng.2017.08.021
$P_{id}9$	DOI: 10.1080/0951192X.2018.1535195
$P_{id}10$	DOI: 10.1109/ACCESS.2019.2962232
$P_{id}11$	DOI: 10.1016/j.procir.2015.12.081
$P_{id}12$	DOI: 10.3233/JIFS171379
$P_{id}13$	DOI: 10.1109/LRA.2019.2931179
$P_{id}14$	DOI: 10.1109/ACCESS.2019.2927743
$P_{id}15$	doi: 10.1109/LRA.2020.2976321
$P_{id}16$	DOI: 10.3390/robotics7030047
$P_{id}17$	DOI: 10.1109/ACCESS.2016.2574979
$P_{id}18$	DOI: 10.1016/j.compeleceng.2017.05.025
$P_{id}19$	DOI: 10.1016/j.heliyon.2020.e03859
$P_{id}20$	DOI: 10.1007/978-3-030-61218-4_1
$P_{id}21$	DOI: 10.1007/978-981-13-2212-9_1
$P_{id}22$	DOI: 10.1109/MNET.2012.6201212
$P_{id}23$	DOI: 10.1007/978-3-319-44350-8_10
$P_{id}24$	DOI: 10.1109/ICONSTEM.2016.7560952
$P_{id}25$	DOI: 10.1016/j.compeleceng.2017.05.024
$P_{id}26$	DOI: 10.1007/978-3-319-94277-3_33
$P_{id}27$	DOI: 10.1109/TVT.2019.2901761
$P_{id}28$	DOI: 10.1109/ACCESS.2016.2574979
$P_{id}29$	DOI: 10.1109/IECON.2011.6119642
$P_{id}30$	DOI: 10.1109/CLOUD.2014.104
$P_{id}31$	DOI: 10.1109/IROS.2013.6697184
$P_{id}32$	DOI: 10.1109/COMST.2021.3061435.
$P_{id}33$	DOI: 10.1177/0142331216687817
$P_{id}34$	DOI: 10.1007/978-3-030-19648-6_61
$P_{id}35$	DOI: 10.1016/j.mechatronics.2017.04.005
$P_{id}36$	DOI: 10.1109/GLOCOM.2016.7841487
$P_{id}37$	DOI: 10.1016/j.ifacol.2016.11.124
$P_{id}38$	DOI: 10.1109/SECON.2018.8479128
$P_{id}39$	DOI: 10.1108/ir-10-2016-0265
Pid40	DOI: 10.1016/j.robot.2019.06.003

Appendix B: Results of the quality assessment scores of the selected 40 primary studies

Paper ID	Paper ID	QA-1	QA-2	QA-3	QA-4	Score
$P_{id}1$	S1	1	0.5	0	1	2.5
$P_{id}2$	S2	1	0.5	0	0.5	2
$P_{id}3$	S3	1	0.5	0	1	2.5
$P_{id}4$	S4	1	0.5	0	1	2.5
$P_{id}5$	S5	1	0.5	1	1	3.5
$P_{id}6$	S6	1	0.5	0	1	3
$P_{id}7$	S7	0.5	1	0	1	2.5
$P_{id}8$	S8	0.5	1	0	1	2.5
$P_{id}9$	S9	1	1	0	1	3
$P_{id}10$	S10	0.5	0.5	1	1	3
$P_{id}11$	S11	0.5	1	0	1	2.5
$P_{id}12$	S12	0.5	1	0	1	2.5
$P_{id}13$	S13	0.5	1	0	1	2.5
$P_{id}14$	S14	1	1	0	1	3
$P_{id}15$	S15	1	1	0	1	3
$P_{id}16$	S16	0.5	1	0	1	2.5
$P_{id}17$	S17	0.5	1	0	1	2.5
$P_{id}18$	S18	0.5	1	1	1	3.5
$P_{id}19$	S19	1	1	0	1	3
$P_{id}20$	S20	1	1	**0**	1	3
$P_{id}21$	S21	1	1	0	1	3
$P_{id}22$	S22	0.5	1	0	1	2.5
$P_{id}23$	S23	1	1	0	1	3
$P_{id}24$	S24	0.5	1	0	1	2.5
$P_{id}25$	S25	0.5	1	0	1	2.5
$P_{id}26$	S26	0.5	1	1	1	3.5
$P_{id}27$	S27	0.5	1	1	1	3.5
$P_{id}28$	S28	1	1	0	1	3
$P_{id}29$	S29	1	1	0	1	3
$P_{id}30$	S30	0.5	1	0	1	2.5
$P_{id}31$	S31	1	1	0	1	3
$P_{id}32$	S32	1	1	0	1	3
$P_{id}33$	S33	1	1	0	1	3
$P_{id}34$	S34	1	0.5	1	1	3.5
$P_{id}35$	S35	1	0.5	1	1	3.5
$P_{id}36$	S36	1	0.5	0	1	2.5
$P_{id}37$	S37	1	0.5	0	1	3
$P_{id}38$	S38	0.5	0.5	0	1	2
$P_{id}39$	S39	0.5	1	0	1	2.5
Pid40	S40	0.5	1	1	1	3.5

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